Literature Status: Same-Szlenk Basis Embeddings

Automatic Math Discovery Run `fa_banach_001`

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Status

This is a **literature_already_answered** packet for the basis-embedding subquestion in arXiv:0706.0656. It records an already-known answer from a later paper, not a new proof from the run.

Source Question

The source paper is:

> Edward Odell, Thomas Schlumprecht, and Andras Zsak, *Banach spaces of bounded Szlenk index*, arXiv:0706.0656.

In the open-problems paragraph on page 4 of the source PDF, the authors ask:

> Is it true that, given $\alpha \leq \omega_1$, every space in $\mathcal{C}_{\omega^\alpha}$ embeds into a space with a basis of the same class?

Here $\mathcal{C}_\xi$ denotes the class of separable reflexive Banach spaces X such that both $\mathrm{Sz}(X)$ and $\mathrm{Sz}(X^*)$ are bounded by ξ.

Supporting Answer

The supporting paper is:

> Edward Odell and Thomas Schlumprecht, *On Zippin's Embedding Theorem of Banach spaces into Banach spaces with bases*, arXiv:1408.3311.

The Main Theorem of the supporting paper, on page 2, states that if X has separable dual, then X embeds into a space W with a shrinking basis such that

$$\mathrm{Sz}(W) = \mathrm{Sz}(X).$$

It also states that if X is reflexive, then W is reflexive and

$$\mathrm{Sz}(W^*) = \mathrm{Sz}(X^*).$$

The same page says that these parts of the Main Theorem answer a question posed by Pełczyński and sharpen the earlier Odell–Schlumprecht–Zsak bounded-Szlenk embedding results.

Identification

Let $X \in \mathcal{C}_{\omega^\alpha}$. Then X is separable and reflexive, and

$$\mathrm{Sz}(X) \leq \omega^\alpha, \qquad \mathrm{Sz}(X^*) \leq \omega^\alpha.$$

Applying the supporting Main Theorem gives a reflexive space W with a basis such that X embeds into W, $\mathrm{Sz}(W) = \mathrm{Sz}(X)$, and $\mathrm{Sz}(W^*) = \mathrm{Sz}(X^*)$. Hence $W \in \mathcal{C}_{\omega^\alpha}$. This is exactly the same-class basis embedding requested in the source question.

The supporting authors explicitly frame their theorem as answering the Pełczyński embedding question and as sharpening the earlier bounded Szlenk-index results, so this packet is classified as an already-known later answer.

Scope Limitations

This packet answers only the first sentence of the second source problem: the existence of a same-class basis superspace for each individual $X \in \mathcal{C}_{\omega^\alpha}$. It does not settle the source paper's universal-element questions for $\mathcal{C}_{\omega^{\alpha\cdot\omega}}$ or for $\mathcal{C}_{\omega^\alpha}$.

Search Evidence

Cheap run indexes for arXiv:0706.0656, bounded Szlenk index, same-class basis embeddings, and Szlenk universality had no prior exact packet. Local later-source search found arXiv:1408.3311 and its Main Theorem. External web search on June 28, 2026 for the exact Szlenk-universality phrases returned arXiv:0706.0656 and the related sequel arXiv:0809.3626, but no newer contradiction to the arXiv:1408.3311 same-Szlenk basis theorem.

References

[1] E. Odell, T. Schlumprecht, and A. Zsak, *Banach spaces of bounded Szlenk index*, arXiv:0706.0656.

[2] E. Odell and T. Schlumprecht, *On Zippin's Embedding Theorem of Banach spaces into Banach spaces with bases*, arXiv:1408.3311.